Dystric Cambisols

Description:

Dystric Cambisols are weakly developed mineral soils with a cambic (Bw) horizon and low base saturation (<50%). The term *Dystric* indicates their low content of exchangeable bases and high acidity, resulting from strong leaching under humid climatic conditions. They typically form from acidic parent materials such as granite, quartzite, or sandstone and are common in humid to sub-humid highlands, especially under forest vegetation. While they have a stable structure and moderate depth, they are generally of low natural fertility unless carefully managed.

Drainage:

- **Wet season:** Generally well-drained; high rainfall leads to leaching and base loss.
- **Dry season:** Retains moderate moisture but dries out quickly in shallow profiles; minimal risk of waterlogging except in depressions.
- **Landscape:** Found on hilly and upland terrain, often with moderate to steep slopes derived from silicate-rich parent rocks or colluvial deposits.

Depth:

- Moderately deep to deep (50–150 cm) depending on topography and parent material.
- Rooting depth is generally good but may be limited by stones, compaction, or hard rock layers in steep landscapes.

Workability:

- When wet, the soil is friable and easy to till, though caution is needed to prevent compaction.
- When dry, it becomes firm and slightly hard, requiring timely tillage to preserve structure.
- Although it has good physical structure and drainage, erosion can be severe on exposed slopes.

Nutrient Richness and Cation Exchange Capacity (CEC):

- CEC ranges from low to moderate (8–20 cmol(+)/kg), depending on clay mineral type and organic matter.
- Base saturation is low (<50%) due to leaching of calcium, magnesium, and potassium.
- Nitrogen and phosphorus are commonly deficient, while potassium and calcium are depleted in deeper layers.
- Organic matter content is moderate (1.0–2.5%) in surface horizons but declines sharply with depth.
- Micronutrient deficiencies (e.g., boron, zinc, molybdenum) may occur under continuous cultivation.

pH:
Strongly to moderately acidic, typically between 4.5 and 5.5. Liming is necessary for most crops to reduce acidity and improve nutrient availability.

Management:

- Apply lime periodically to raise pH and base saturation.
- Incorporate organic matter (manure, compost, green manure) to improve fertility, structure, and water-holding capacity.
- Use balanced fertilizer applications, emphasizing nitrogen and phosphorus; supplement with micronutrients when necessary.
- Employ soil conservation measures such as contour plowing, terracing, and mulching to minimize erosion.
- Maintain vegetative cover or use agroforestry to stabilize soil and enhance biological activity.

Suitable Crops:

- After liming and fertility improvement, Dystric Cambisols can support a range of crops including maize, beans, potatoes, tea, coffee, bananas, and pyrethrum.
- In less acidic conditions, they can also support forestry species such as grevillea, eucalyptus, and cypress.